

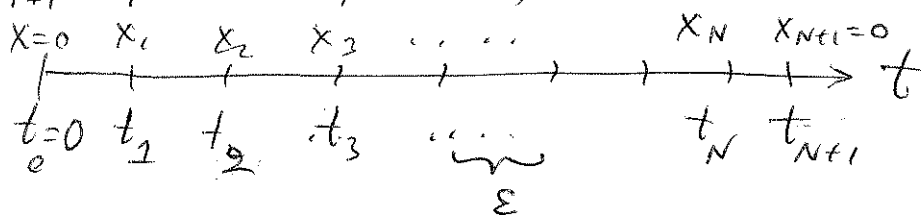
1.1.4 Examples

Example 1.1

$$W(x, t | 0, 0) = \int_{C[0, 0; x, t]} d_w x(\tau) = \frac{1}{\sqrt{4\pi\Delta t}} e^{-x^2/4\Delta t}$$

Let us calculate this path integral explicitly for $x=0$, or for $C[0, 0; 0, t]$. It should be $= 1/\sqrt{4\pi\Delta t}$.

$$t_{i+1} - t_i = \varepsilon = t/(N+1)$$



of interval
= (N+1)

$$\int_{C[0, 0; 0, t]} d_w x(\tau) = \lim_{\substack{\varepsilon \rightarrow 0 \\ N \rightarrow \infty}} \int_{-\infty}^{\infty} dx_1 \frac{e^{-x_1^2/4\Delta t_1}}{\sqrt{4\pi\Delta t_1}} \times \int_{-\infty}^{\infty} dx_2 \frac{e^{-(x_1-x_2)^2/4\Delta(t_2-t_1)}}{\sqrt{4\pi\Delta(t_2-t_1)}} \times$$

$$x_1 \dots x_N \int dx_N \frac{e^{-(x_N-x_{N-1})^2/4\Delta(t_N-t_{N-1})}}{\sqrt{4\pi\Delta(t_N-t_{N-1})}} \dots \frac{e^{-(x_{N+1}-x_N)^2/4\Delta(t_{N+1}-t_N)}}{\sqrt{4\pi\Delta(t_{N+1}-t_N)}}$$

$$= \lim_{\substack{\varepsilon \rightarrow 0 \\ N \rightarrow \infty}} \frac{1}{(4\pi\Delta\varepsilon)^{\frac{N+1}{2}}} \int dx_1 \int dx_2 \dots \int dx_N \exp \left[-\frac{1}{4\Delta\varepsilon} \sum_{i=0}^{N+1} (x_{i+1} - x_i)^2 \right]$$

where $x_0=0$ and $x_{N+1}=0$ as well.

The trick is in writing

$$\sum_{i=0}^{N+1} (x_{i+1} - x_i)^2 = \sum_{i,j=0}^{N+1} x_i A_{ij} x_j$$

with $A_{ii} = 2$ and $A_{i,i+1} = A_{i+1,i} = -1$

$$\sum_{i=0}^{N+1} (x_{i+1} - x_i)^2 = \sum_{i=0}^{N+1} (x_{i+1}^2 + x_i^2 - 2x_{i+1}x_i)$$

$$= \left(\underbrace{x_1^2 + x_2^2 + \dots + x_{N+1}^2}_0 \right) + \left(\underbrace{x_0^2 + x_1^2 + \dots + x_N^2}_0 \right) - 2 \left(\underbrace{x_1^2}_{0} x_0 + \underbrace{x_2^2}_{0} x_1 + \dots + \underbrace{x_{N+1}^2}_{0} x_N \right)$$

$$= 2(x_1^2 + \dots + x_N^2) - 2(x_2 x_1 + x_3 x_2 + \dots + x_N x_{N-1})$$

r.e.

$$A_N = (N \times N \text{ matrix}) = \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & \ddots \\ & & \ddots & 2 \end{pmatrix} \quad (1.1.84)$$

$$\hookrightarrow \lim_{\substack{\epsilon \rightarrow 0 \\ N \rightarrow \infty}} \left(\frac{1}{\sqrt{4\pi\epsilon}} \right)^{N+1} \int dx_1 dx_2 \dots dx_N \exp \left[-\frac{1}{4\epsilon} \sum_{i,j=1}^N x_i A_{ij} x_j \right]$$

$$\frac{(\sqrt{\pi})^N}{\sqrt{\det\left(\frac{A}{4\epsilon}\right)}} = \frac{(\sqrt{\pi})^N}{\left(\frac{1}{\sqrt{4\epsilon}}\right)^N \sqrt{\det A_N}}$$

$$= \lim_{\substack{\epsilon \rightarrow 0 \\ N \rightarrow \infty}} \frac{1}{(\sqrt{4\pi\epsilon})^{N+1}} \cdot \frac{(\sqrt{4\pi\epsilon})^N}{\sqrt{\det A_N}} = \lim_{\substack{\epsilon \rightarrow 0 \\ N \rightarrow \infty}} \frac{1}{\sqrt{4\pi\epsilon} \det A_N}$$

We need to calculate $\det A_N$. ~~We prove th~~

$$N=1 \rightarrow \det A_1 = 2$$

$$N=2 \rightarrow \det A_2 = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 4 - 1 = 3$$

$$N=3 \rightarrow \det A_3 = \begin{vmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{vmatrix} = 2(4-1) + 1(-2) = 6-2=4$$

it looks like $\det A_N = N+1$. Expanding $\det A_N$ w.r.t the ~~1st~~ 1st column:

$$|A_N| = \begin{vmatrix} 2 & -1 & & 0 \\ -1 & 2 & & \\ & & \ddots & -1 \\ 0 & -1 & 2 & \end{vmatrix} = 2 \underbrace{\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}}_{A_{(N-1)}} - (-1) \begin{vmatrix} 2 & -1 & 0 & 0 & \dots \\ -1 & 2 & -1 & 0 & \dots \\ 0 & -1 & 2 & -1 & 0 & \dots \\ 0 & 0 & -1 & 2 & -1 & 0 & \dots \\ \vdots & & & & & \ddots \end{vmatrix}$$

$$= 2|A_{N-1}| + \begin{vmatrix} -1 & 0 & 0 & 0 & \dots \\ -1 & 2 & -1 & 0 & \dots \\ 0 & -1 & 2 & -1 & 0 & \dots \end{vmatrix} = \left(\begin{array}{l} \text{open it w.r.t} \\ \text{1st row} \end{array} \right) =$$

$$= 2 |A_{N-1}| + (-1) \underbrace{\begin{vmatrix} 2 & -1 & 0 & \dots \\ -1 & 2 & -1 & \dots \\ 0 & -1 & 2 & -1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{vmatrix}}_{A_{N-2}} \equiv 2 |A_{N-1}| - |A_{N-2}|$$

Now we can check the assumption:

$$\det A_{N+1} = 2 \det A_N - \det A_{N-1} = 2(N+1) - N = 2N+2-N = N+2, \text{ as required.}$$

$$\hookrightarrow \int_{C[0,0;0,t]} d_w X(\tau) = \lim_{\substack{\epsilon \rightarrow 0 \\ N \rightarrow \infty}} \frac{1}{\sqrt{4\pi\Delta} \underbrace{\epsilon(N+1)}_t} \equiv \frac{1}{\sqrt{4\pi\Delta} t}.$$

Example 1.0 Use integration one-by-one, this time for a general $0,0 \rightarrow x,t$ trajectories set:

$$\int_{C[0,0;x,t]} d_w X(\tau) = \lim_{\substack{\epsilon \rightarrow 0 \\ N \rightarrow \infty}} \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 \dots \int_{-\infty}^{\infty} dx_N \frac{1}{(\sqrt{4\pi\Delta}\epsilon)^{N+1}} \times \\ \times \exp \left[-\frac{1}{4\Delta\epsilon} \sum_{i=0}^N (x_{i+1} - x_i)^2 \right]$$

This time, $x_0=0$, but $x_{N+1} \neq 0$, $x_{N+1}=x$. We should get

$$\int_{C[0,0;x,t]} d_w X(\tau) = \frac{1}{\sqrt{4\pi\Delta} t} e^{-x^2/4\Delta t}$$

We use the following identity:

$$\int_{-\infty}^{\infty} dx \exp[-a(x-y)^2 - b(x-z)^2] = \sqrt{\frac{\pi}{a+b}} \exp\left[-\frac{ab}{a+b}(y-z)^2\right]$$

But first we change variables:

$$y_i = x_i / \sqrt{4\Delta\epsilon} \rightarrow \frac{1}{4\Delta\epsilon} (x_{i+1} - x_i)^2 = (y_{i+1} - y_i)^2$$

$$\int_C = \frac{1}{(\sqrt{4\pi\Delta\varepsilon})^{N+1}} (\sqrt{4\pi\Delta\varepsilon})^N \int dy_1 \int dy_2 \dots \int dy_N \exp\left[-\sum_{i=0}^N (y_{i+1} - y_i)^2\right]$$

$$= \frac{1}{\sqrt{4\pi\Delta\varepsilon} (\sqrt{\pi})^{N+1}} \int dy_1 \dots$$

Here:

$$\int dy_N \exp\left[-(y_{N+1} - y_N)^2 - (y_N - y_{N-1})^2\right] = \sqrt{\frac{\pi}{1+1}} \exp\left[-\frac{1}{2} (y_{N+1} - y_{N-1})^2\right]$$

$$= \sqrt{\frac{\pi}{2}} \exp\left[-\frac{1}{2} (y - y_{N-1})^2\right], \quad y = y_{N+1}$$

$$\int dy_{N-1} \exp\left[-\frac{1}{2} (y - y_{N-1})^2 - (y_{N-1} - y_{N-2})^2\right] =$$

$$= \sqrt{\frac{\pi}{1+\frac{1}{2}}} \exp\left[-\frac{\frac{1}{2}}{1+\frac{1}{2}} (y - y_{N-2})^2\right] = \sqrt{\frac{\pi}{\frac{3}{2}}} \exp\left[-\frac{1}{3} (y - y_{N-2})^2\right]$$

$$\int dy_{N-2} \exp\left[-\frac{1}{3} (y - y_{N-2})^2 - (y_{N-2} - y_{N-3})^2\right] =$$

$$= \sqrt{\frac{\pi}{1+\frac{1}{3}}} \exp\left[-\frac{\frac{1}{3}}{1+\frac{1}{3}} (y - y_{N-3})^2\right] = \sqrt{\frac{\pi}{\frac{4}{3}}} \exp\left[-\frac{1}{4} (y - y_{N-3})^2\right]$$

$$\int dy_{N-i} \exp\left[-\frac{1}{i+1} (y - y_{N-i})^2 - (y_{N-i} - y_{N-i-1})^2\right] =$$

$$= \sqrt{\frac{\pi}{1+\frac{1}{i+1}}} \exp\left[-\frac{\frac{1}{i+1}}{1+\frac{1}{i+1}} (y - y_{N-i-1})^2\right] = \sqrt{\frac{\pi}{\frac{i+2}{i+1}}} \exp\left[-\frac{1}{i+2} (y - y_{N-i-1})^2\right]$$

etc.

$$\hookrightarrow \int_{C[0,0;x,t]} d_W x[\tau] = \frac{1}{\sqrt{4\pi\Delta\varepsilon}} \frac{1}{(\sqrt{\pi})^{N+1}} \cdot \frac{(\sqrt{\pi})^N}{\sqrt{2 \times \frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \dots \times \frac{N+1}{N}}}$$

$$x \exp \left[-\frac{1}{N+1} (y - y_0)^2 \right] = \frac{1}{\sqrt{4\pi\Delta\varepsilon(N+1)}} e^{-\frac{y^2}{N+1}}$$

$$\rightarrow \frac{1}{\sqrt{4\pi\Delta\varepsilon}} e^{-\frac{x^2}{4\Delta\varepsilon(N+1)}} \quad \text{O.E.D.}$$

Example 1.2 $F[x(\tau)] = x(s)$, s - a fixed time $\in [0, t]$.

$$\int_{C[x_0, 0; t]} d_W x(\tau) \cdot x(s) = \int dx_s \cdot x_s \underbrace{\int_{C[x_0, 0; x_s, s]} d_W x(\tau) \cdot x(s)}_{\frac{1}{\sqrt{4\pi\Delta\varepsilon}} e^{-\frac{(x_s - x_0)^2}{4\Delta\varepsilon}}}$$

$$\times \underbrace{\int_{C[x_s, s; t]} d_W x(\tau)}_{=1} = \frac{1}{\sqrt{4\pi\Delta\varepsilon}} \int_{-\infty}^{\infty} x_s e^{-\frac{(x_s - x_0)^2}{4\Delta\varepsilon}} dx_s$$

$$= \frac{1}{\sqrt{4\pi\Delta\varepsilon}} \int \left[\underbrace{(x_s - x_0)}_0 + \underbrace{x_0}_{\text{normalisation}} \right] e^{-\frac{(x_s - x_0)^2}{4\Delta\varepsilon}} dx_s = x_0.$$

• Another method via discretisation: $\varepsilon = t_{i+1} - t_i$, $x_s \rightarrow x_e$

$$\int_{C[x_0, 0; t]} d_W x(\tau) x(s) \Rightarrow \int dx_1 \dots \int dx_e \dots \int dx_N \cdot x_e \frac{1}{(4\pi\Delta\varepsilon)^{N/2}}$$

$$\times \exp \left[-\frac{1}{4\Delta\varepsilon} \sum_{i=1}^N (x_i - x_{i-1})^2 \right] =$$

$$= \frac{1}{(\sqrt{4\pi\epsilon})^N} \times \left(\int dx_1 \dots \int dx_{e-1} \right) \cdot \int dx_e \cdot x_e \left(\int dx_{e+1} \dots dx_N \right) \times$$

$$\times \exp \left[-\frac{1}{4\epsilon} \sum_{i=1}^{e-1} (x_i - x_{i-1})^2 \right] e^{-\frac{1}{4\epsilon} (x_e - x_{e-1})^2} \times \int dx$$

$$\times \exp \left[-\frac{1}{4\epsilon} \sum_{i=e+1}^N (x_i - x_{i-1})^2 \right]$$

Here

$$\frac{1}{(\sqrt{4\pi\epsilon})^N} \int dx_{e+1} \dots dx_N \dots =$$

$$= \prod_{i=e+1}^N \int_{-\infty}^{\infty} \frac{dx_i}{\sqrt{4\pi\epsilon}} e^{-\frac{1}{4\epsilon} (x_i - x_{i-1})^2} = \frac{1}{\sqrt{4\pi\epsilon}} \sqrt{\frac{\pi}{1/4\epsilon}} = 1$$

(this of course corresponds to $\int_C d_w X(\tau) = 1$)

Integrals

$$\prod_{i=1}^{e-1} \int_{-\infty}^{\infty} \frac{dx_i}{\sqrt{4\pi\epsilon}} e^{-\frac{1}{4\epsilon} (x_i - x_{i-1})^2} \times \left(e^{-\frac{1}{4\epsilon} (x_e - x_{e-1})^2} \right) \frac{1}{\sqrt{4\pi\epsilon}}$$

corresponds to

$$\int_C \frac{1}{\sqrt{4\pi\epsilon}} e^{-\frac{(x_e - x_0)^2}{4\epsilon}} =$$

which is checked by direct calculation:

$$\int_C [x_0, 0; x_e, t_e] = \frac{1}{\sqrt{4\pi\epsilon t_e}} e^{-\frac{(x_e - x_0)^2}{4\epsilon t_e}}$$

so that

$$\int_C [x_0, 0; *] d_w X(\tau) \cdot X(s) = \int_{-\infty}^{\infty} dx_s \frac{x_s}{\sqrt{4\pi\epsilon t_e}} e^{-\frac{(x_e - x_0)^2}{4\epsilon t_e}} = x_0$$

again.

Example 1.3

$$I_3 = \int_{C[0,0;t]} d_w x(\tau) F[x(\tau)]$$

$$\text{with } F[x(\tau)] = F\left(\int_0^t a(\tau) x(\tau) d\tau\right)$$

where $F(x), a(x)$ - some functions.

$$I_3 = \int \frac{dx_1}{\sqrt{4\pi\Delta\Delta t_1}} \int \frac{dx_2}{\sqrt{4\pi\Delta\Delta t_2}} \dots \int \frac{dx_N}{\sqrt{4\pi\Delta\Delta t_N}} F\left(\sum_{i=1}^N \underbrace{a(t_i)}_{q_i} \underbrace{x(t_i)}_{x_i} \Delta t_i\right) \times \\ \times \exp\left(-\frac{1}{4\Delta} \sum_{i=1}^N \frac{(x_i - x_{i-1})^2}{\Delta t_i}\right)$$

Change variables: $y_i = x_i - x_{i-1}$, $i = 1, \dots, N$

To calculate the Jacobian:

$$x_1 = y_1 \quad (x_0 = 0)$$

$$x_2 = y_1 + y_2$$

$$x_3 = y_1 + y_2 + y_3 \dots$$

$$\left| \frac{\partial x}{\partial y} \right| = \begin{vmatrix} \partial x_1 / \partial y_1 & \partial x_1 / \partial y_2 & \dots \\ \partial x_2 / \partial y_1 & \partial x_2 / \partial y_2 & \dots \\ \partial x_3 / \partial y_1 & \partial x_3 / \partial y_2 & \partial x_3 / \partial y_3 \dots \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{vmatrix} = 1$$

as the det of a Δ matrix = product of diagonal elements.

$$\hookrightarrow I_3 = \int \frac{dy_1}{\sqrt{4\pi\Delta\Delta t_1}} \dots \int \frac{dy_N}{\sqrt{4\pi\Delta\Delta t_N}} F\left(\sum_{i=1}^N y_i q_i \Delta t_i\right) \exp\left(-\frac{1}{4\Delta} \sum_{i=1}^N \frac{y_i^2}{\Delta t_i}\right)$$

where

$$\left(\sum_{i=1}^N y_i q_i \Delta t_i\right) = q_1 x_1 \Delta t_1 + q_2 x_2 \Delta t_2 + \dots = q_1 \Delta t_1 (y_1) + q_2 (y_1 + y_2) \Delta t_2 + \dots \\ = y_1 (q_1 \Delta t_1 + q_2 \Delta t_2 + \dots + q_N \Delta t_N) + y_2 (q_2 \Delta t_2 + \dots + q_N \Delta t_N) + \dots$$

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$$+ y_N(a_N \Delta t_N) = \sum_{i=1}^N y_i A_i,$$

$$A_i = \sum_{j=i}^N a_j \Delta t_j \quad 1 \leq i \leq N$$

New variable again:

$$z_i = A_i y_i \rightarrow y_i = A_i^{-1} z_i$$

$$\left| \frac{\partial y}{\partial z} \right| = \prod_{i=1}^N A_i^{-1} = \prod_{i=1}^N \frac{1}{\sqrt{A_i^2}}$$

$$\hookrightarrow I_3 = \int \frac{dz_1}{\sqrt{4\pi\Delta A_1^2\Delta t_1}} \times \dots \times \int \frac{dz_N}{\sqrt{4\pi\Delta A_N^2\Delta t_N}} \cdot F(z_1 + \dots + z_N) \exp\left[-\frac{1}{4\pi} \sum_{i=1}^N \frac{z_i^2}{A_i^2\Delta t_i}\right]$$

Next step, we introduce new variables:

$$y = z_1 + z_2, \quad \xi = z_2 \text{ instead of } z_1, z_2.$$

$$\begin{aligned} z_2 &= \xi \\ z_1 &= y - \xi \end{aligned} \quad , \quad \frac{\partial(z_1, z_2)}{\partial(y, \xi)} = \begin{vmatrix} 1 & -1 \\ 0 & 1 \end{vmatrix} = 1$$

$$\int \frac{dz_1}{\sqrt{4\pi\Delta A_1^2\Delta t_1}} \cdot \int \frac{dz_2}{\sqrt{4\pi\Delta A_2^2\Delta t_2}} \rightarrow \int \frac{dy}{\sqrt{\dots}} \int \frac{d\xi}{\sqrt{\dots}} \int \frac{dz_3}{\sqrt{\dots}} \dots \int \frac{dz_N}{\sqrt{\dots}}$$

$$\times F(y + z_3 + z_4 + \dots + z_N) \exp\left[-\frac{1}{4\pi} \left(\frac{(y-\xi)^2}{A_1^2\Delta t_1} + \frac{\xi^2}{A_2^2\Delta t_2} \right) - \frac{1}{4\pi} \sum_{i=3}^N \frac{z_i^2}{A_i^2\Delta t_i}\right]$$

Now we can perform integration over ξ since F does not depend on it:

$$\int \frac{d\xi}{\sqrt{\dots}} e^{-\frac{1}{4\pi\Delta A_1^2\Delta t_1} (y-\xi)^2} e^{-\frac{1}{4\pi\Delta A_2^2\Delta t_2} \xi^2} =$$

$$= \sqrt{\frac{\pi}{\frac{1}{4\pi\Delta A_1^2\Delta t_1} + \frac{1}{4\pi\Delta A_2^2\Delta t_2}}} \exp\left\{-\frac{\frac{1}{4\pi\Delta A_1^2\Delta t_1} \cdot \frac{1}{4\pi\Delta A_2^2\Delta t_2}}{\frac{1}{4\pi\Delta A_1^2\Delta t_1} + \frac{1}{4\pi\Delta A_2^2\Delta t_2}} y^2\right\}$$

$$= \frac{1}{\sqrt{4\pi(A_1^2\Delta t_1 + A_2^2\Delta t_2)}} \exp\left\{-\frac{y^2}{4\Delta(A_1^2\Delta t_1 + A_2^2\Delta t_2)}\right\} \times$$

$$\times \frac{\sqrt{\pi} \cdot \sqrt{4\Delta A_1^2\Delta t_1} \cdot \sqrt{4\Delta A_2^2\Delta t_2}}{\sqrt{4\Delta(A_1^2\Delta t_1 + A_2^2\Delta t_2)}}$$

Incorporating also prefactors $1/\sqrt{4\Delta A_1^2\Delta t_1} \cdot 1/\sqrt{4\Delta A_2^2\Delta t_2}$ from the $\int dz_1 \int dz_2$, we obtain:

$$\int dz_1 \int dz_2 \rightarrow \frac{1}{\sqrt{4\Delta\pi(A_1^2\Delta t_1 + A_2^2\Delta t_2)}} \int \exp\left\{-\frac{y^2}{4\Delta(A_1^2\Delta t_1 + A_2^2\Delta t_2)}\right\} dy$$

$$\begin{aligned} I_3 = & \int \frac{dz}{\sqrt{4\Delta\pi(A_1^2\Delta t_1 + A_2^2\Delta t_2)}} \int \frac{dz_3}{\sqrt{4\Delta\pi A_3^2\Delta t_3}} \dots \int \frac{dz_N}{\sqrt{4\Delta\pi A_N^2\Delta t_N}} \times \\ & \times F(y + z_3 + \dots + z_N) \exp\left\{-\frac{y^2}{4\Delta(A_1^2\Delta t_1 + A_2^2\Delta t_2)} - \frac{1}{4\Delta} \sum_{i=3}^N \frac{z_i^2}{A_i^2\Delta t_i}\right\} \end{aligned}$$

Now we introduce new variables:

$$\lambda = y + z_3, \quad \sigma = z_3 \rightarrow \text{Jacobian} = 1$$

$$\begin{aligned} I_3 = & \int \frac{d\lambda}{\sqrt{4\Delta\pi(A_1^2\Delta t_1 + A_2^2\Delta t_2)}} \int \frac{d\sigma}{\sqrt{4\Delta\pi A_3^2\Delta t_3}} \int \frac{dz_4}{\sqrt{\pi}} \dots F(\lambda + z_4 + \dots + z_N) \times \\ & \times \exp\left\{-\frac{(\lambda - \sigma)^2}{4\Delta(A_1^2\Delta t_1 + A_2^2\Delta t_2)} - \frac{\sigma^2}{4\Delta A_3^2\Delta t_3} - \frac{1}{4\Delta} \sum_{i=4}^N \frac{z_i^2}{A_i^2\Delta t_i}\right\} \end{aligned}$$

Again we integrate over σ :

$$\int \frac{d\sigma}{\sqrt{4\Delta\pi A_3^2\Delta t_3}} \exp\left\{-\frac{(\lambda - \sigma)^2}{4\Delta(A_1^2\Delta t_1 + A_2^2\Delta t_2)} - \frac{\sigma^2}{4\Delta A_3^2\Delta t_3}\right\} =$$

$$= \frac{1}{\sqrt{4\pi\omega A_3^2 \Delta t_3}} \sqrt{\frac{1}{4\omega(A_1^2 \Delta t_1 + A_2^2 \Delta t_2)} + \frac{1}{4\omega A_3^2 \Delta t_3}} \times$$

$$\times \exp \left\{ - \frac{\frac{1}{4\omega(A_1^2 \Delta t_1 + A_2^2 \Delta t_2)} + \frac{1}{4\omega A_3^2 \Delta t_3}}{\frac{1}{4\omega(A_1^2 \Delta t_1 + A_2^2 \Delta t_2)} + \frac{1}{4\omega A_3^2 \Delta t_3}} \cdot \lambda^2 \right\}$$

$$= \frac{\sqrt{4\omega(A_1^2 \Delta t_1 + A_2^2 \Delta t_2)}}{\sqrt{4\omega(A_1^2 \Delta t_1 + A_2^2 \Delta t_2 + A_3^2 \Delta t_3)}} \exp \left\{ - \frac{\lambda^2}{4\omega(A_1^2 \Delta t_1 + A_2^2 \Delta t_2 + A_3^2 \Delta t_3)} \right\}$$

so that

$$\int d\lambda \int d\omega \rightarrow \int \frac{d\lambda}{\sqrt{4\pi\omega(A_1^2 \Delta t_1 + A_2^2 \Delta t_2 + A_3^2 \Delta t_3)}} e^{-\frac{\lambda^2}{4\omega(A_1^2 \Delta t_1 + \dots + A_3^2 \Delta t_3)}}$$

Continuing this process, we have just the last integration left:

$$I_3 = \int \frac{d\xi}{\sqrt{4\pi\omega \sum_{i=1}^N A_i^2 \Delta t_i}} F(\xi) \exp \left(- \frac{\xi^2}{4\omega \sum_{i=1}^N A_i^2 \Delta t_i} \right)$$

$$\xrightarrow[N \rightarrow \infty, \Delta t_i \rightarrow 0]{} \int_{-\infty}^{\infty} \frac{d\xi}{\sqrt{4\pi\omega R_t}} F(\xi) \exp \left(- \frac{\xi^2}{4\omega R_t} \right),$$

where

$$R_t \equiv \lim_{\Delta t_i \rightarrow 0} \sum_{i=1}^N A_i^2 \Delta t_i = \lim_{\Delta t_i \rightarrow 0} \sum_{i=1}^N \left(\sum_{j=i}^N a_j \Delta t_j \right)^2 \Delta t_i$$

$$= \int_0^t dt \left(\int_{\tau}^t d\tau' a(\tau') \right)^2.$$

Example 1.4

$$I_4 = \int_{C[0,0;t]} d_W x(\tau) \exp \left\{ - \int_0^t d\tau \rho(\tau) x^2(\tau) \right\}$$

for simplicity

Using discretisation: $x_1, \dots, x_N$; $\Delta t_i = \varepsilon \equiv t/N$ (4) = 1

$$I_4^{(N)} = \int \frac{dx_1}{\sqrt{\pi\varepsilon}} \int \frac{dx_2}{\sqrt{\pi\varepsilon}} \dots \int \frac{dx_N}{\sqrt{\pi\varepsilon}} \exp \left\{ - \sum_{i=1}^N p_i x_i^2 \varepsilon - \sum_{i=1}^N \frac{(x_i - x_{i-1})^2}{\varepsilon} \right\}$$

$$= \int \frac{dx_1}{\sqrt{\pi\varepsilon}} \int \dots \int \frac{dx_N}{\sqrt{\pi\varepsilon}} \exp \left\{ - \sum_{i,j=1}^N a_{ij} x_i x_j \right\}, \quad (1.1.103)$$

with

$$a_{ij} = \begin{cases} \frac{2}{\varepsilon} + p_i \varepsilon \equiv q_i, & i=j \in (1, \dots, N) \\ -1/\varepsilon, & \text{when } i=j \pm 1 \end{cases} \quad \left| \quad q_N \equiv p_N \varepsilon + \frac{1}{\varepsilon} \right.$$

is again a three-diagonal matrix:

$$a = \begin{pmatrix} q_1 & -1/\varepsilon & 0 & 0 & 0 \\ -1/\varepsilon & q_2 & -1/\varepsilon & 0 & 0 \\ 0 & -1/\varepsilon & q_3 & -1/\varepsilon & 0 \\ 0 & 0 & -1/\varepsilon & \ddots & -1/\varepsilon \\ \vdots & \vdots & \vdots & 0 & -1/\varepsilon & q_N \end{pmatrix}$$

$$\hookrightarrow I_4^{(N)} = \frac{1}{(\sqrt{\pi\varepsilon})^N} \frac{(\sqrt{\pi})^N}{\sqrt{\det a}} = \frac{1}{(\sqrt{\varepsilon})^N \sqrt{\det a}} = \frac{1}{\sqrt{\det(\varepsilon a)}},$$

$$\varepsilon a = \begin{pmatrix} \varepsilon q_1 & -1 & & & \\ -1 & \varepsilon q_2 & -1 & & \\ & -1 & \ddots & & \\ & & & \ddots & -1 \\ & & & & \varepsilon q_N \end{pmatrix}$$

(each element a_{ij} is multiplied in εa by ε)

Consider its determinant $\Delta^{(N)}$. To this end, consider its subdeterminant $\Delta_k^{(N)}$ with first $(k-1)$ rows and columns removed:

$$D_k^{(N)} = \begin{vmatrix} \varepsilon a_k & -1 & 0 & 0 & \dots \\ -1 & \varepsilon a_{k+1} & -1 & 0 & \dots \\ 0 & -1 & \varepsilon a_{k+2} & -1 & \dots \\ 0 & 0 & -1 & \dots & \dots \end{vmatrix}$$

Open up along the 1st row;

$$D_k^{(N)} = \varepsilon a_k D_{k+1}^{(N)} - (-1) \begin{vmatrix} \cancel{\varepsilon a_k} & \cancel{-1} & \cancel{0} & \dots \\ 0 & \varepsilon a_{k+1} & -1 & \dots \\ 0 & -1 & \varepsilon a_{k+2} & \dots \end{vmatrix}$$

open along the 1st column

$$= \varepsilon a_k D_{k+1}^{(N)} + \varepsilon a_k \begin{vmatrix} \cancel{\varepsilon a_k} & \cancel{-1} & \cancel{0} & \dots \\ \varepsilon a_{k+1} & -1 & 0 & \dots \\ -1 & \varepsilon a_{k+2} & -1 & \dots \end{vmatrix} + (-1) \begin{vmatrix} \varepsilon a_{k+2} & -1 & \dots \\ -1 & \varepsilon a_{k+3} & -1 & \dots \end{vmatrix}$$

$$= \varepsilon a_k D_{k+1}^{(N)} - \varepsilon a_k D_{k+2}^{(N)}$$

$$\text{or } \frac{D_k^{(N)} - 2D_{k+1}^{(N)} + D_{k+2}^{(N)}}{\varepsilon^2} = \frac{\varepsilon a_k D_{k+1}^{(N)} - 2D_{k+1}^{(N)}}{\varepsilon^2} = \frac{D_k^{(N)}}{D_{k+1}^{(N)}}$$

where

$$\text{RHS} = \frac{1}{\varepsilon^2} [\varepsilon a_k - 2] D_{k+1}^{(N)} = \frac{1}{\varepsilon^2} [p_k \varepsilon^2 + 2 - 2] D_{k+1}^{(N)} = p_k D_{k+1}^{(N)}$$

$$\hookrightarrow \frac{D_k^{(N)} - 2D_{k+1}^{(N)} + D_{k+2}^{(N)}}{\varepsilon^2} = p_k D_{k+1}^{(N)} \quad (1.1.107)$$

Index k corresponds to the time instant $t_k = k\varepsilon$

$$\text{LHS} = \frac{d^2}{dt^2} D(t) \text{ when } N \rightarrow \infty$$

RHS $\rightarrow p(t) D(t)$ in the same limit:

$$\boxed{\frac{d^2 D(t)}{dt^2} = p(t) D(t)} \quad (1.1.108)$$

$$\boxed{D(t) = \lim_{N \rightarrow \infty} D_k^{(N)}}$$

We need $\mathcal{Q}_1^{(N)} \equiv \det(\varepsilon a)^{-\frac{1}{2}}$ which corresponds to $\mathcal{Q}(a)$. 3
 Note:

$$\mathcal{Q}(t) = \lim_{N \rightarrow \infty} \mathcal{Q}_N^{(N)} = \varepsilon q_N = p_N \varepsilon^2 + 1 \xrightarrow{\varepsilon \rightarrow 0} 1$$

$$\begin{aligned} \left. \frac{d\mathcal{Q}(t)}{d\tau} \right|_{t=t} &= \lim_{N \rightarrow \infty} \frac{\mathcal{Q}_N^{(N)} - \mathcal{Q}_{N-1}^{(N)}}{\varepsilon} = \frac{1}{\varepsilon} \left[p_N \varepsilon^2 + 1 - \right. \\ &\quad \left. - \begin{vmatrix} \varepsilon q_{N-1} & -1 \\ -1 & \varepsilon q_N \end{vmatrix} \right] = \frac{1}{\varepsilon} \left[p_N \varepsilon^2 + 1 - \varepsilon^2 q_{N-1} q_N + 1 \right] \\ &= \frac{1}{\varepsilon} \left[2 + p_N \varepsilon^2 - \varepsilon^2 (2 + p_{N-1} \varepsilon^2)(1 + p_N \varepsilon^2) \right] \xrightarrow{\varepsilon \rightarrow 0} \\ &= \frac{1}{\varepsilon} \left[\cancel{2} + p_N \varepsilon^2 - \cancel{2} - p_{N-1} \varepsilon^2 - 2 p_N \varepsilon^2 - p_N p_{N-1} \varepsilon^4 \right] \\ &= -p_N p_{N-1} \varepsilon^3 - (p_{N-1} + p_N) \varepsilon \rightarrow 0 \text{ as } \varepsilon \rightarrow 0 \end{aligned}$$

Therefore, Eq. (1.1.108) is to be solved with the boundary conditions

$$\boxed{\mathcal{Q}(t) = 1, \quad \left. \frac{d\mathcal{Q}}{d\tau} \right|_{t=\tau} = 0} \quad \begin{array}{l} (1.1.109) \\ (1.1.110) \end{array}$$

Then,
$$I_4 = \frac{1}{\sqrt{\mathcal{Q}(0)}} \quad (1.1.111)$$

Example 1.5 - 396 - 4 -

The same as before, but with the conditional Wiener measure:

$$\begin{aligned}
 I_4^{\text{cond}} &= \int_{C[0,0;X,t]} d_W X(t) \exp \left\{ - \int_0^t p(t) X^2(t) dt \right\} \quad (1.1.112) \\
 &= \int \frac{dx_1}{\sqrt{\pi \varepsilon}} \dots \int \frac{dx_{N+1}}{\sqrt{\pi \varepsilon}} \frac{1}{\sqrt{\pi \varepsilon}} \exp \left\{ - \sum_{i=1}^{N+1} p_i X_i^2 \varepsilon - \sum_{i=1}^{N+1} \frac{(X_i - X_{i-1})^2}{\varepsilon} \right\} \\
 &= \frac{1}{\sqrt{\pi \varepsilon}} \int \frac{dx_1 \dots dx_{N+1}}{(\sqrt{\pi \varepsilon})^{N+1}} \exp \left\{ - \sum_{i,j=1}^{N+1} a_{ij} X_i X_j \right\}
 \end{aligned}$$

A = the same matrix as in the previous problem.

However, we need to isolate the $(N+1)$ th term:

$$\sum_{i,j=1}^N a_{ij} X_i X_j + \sum_{i=1}^N \frac{(X_i - X_{i-1})^2}{\varepsilon} + \frac{X_{N-1}^2}{\varepsilon} + \frac{X_N^2}{\varepsilon} + \frac{2}{\varepsilon} X_N X_{N+1} + \dots$$

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1N} & a_{1,N+1} \\ a_{21} & & & & & a_{2,N+1} \\ & & & & & \vdots \\ & & & & a_{NN} & a_{N,N+1} \\ a_{N+1,1} & & & & a_{N+1,N} & a_{N+1,N+1} \end{pmatrix}$$

$$a_{ij} = \delta_{ij} \left(p_i \varepsilon + \frac{2}{\varepsilon} \right) \quad \text{if } i=j \text{ (between 1 and } N+1) \quad (1 \leq i < N+1)$$

$$\frac{1}{\varepsilon} = p_i \varepsilon + \frac{1}{\varepsilon}, \quad i = N+1 \text{ axis}$$

and $-1/\varepsilon$ along the diagonal on both sides:

$$A = \begin{pmatrix} a_1 - 1/\epsilon & & & & \\ -1/\epsilon & a_2 - 1/\epsilon & & & \\ & & \ddots & & \\ & & & -1/\epsilon & a_N - 1/\epsilon \\ & & & -1/\epsilon & a_{N+1} \end{pmatrix}$$

We introduce the δ -function to include x_{N+1} into the integration:

$$\begin{aligned} I_4^{\text{cond}} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} ds \int \frac{dx_1}{\sqrt{4\pi\epsilon}} \dots \int \frac{dx_N}{\sqrt{4\pi\epsilon}} \int \frac{dx_{N+1}}{\sqrt{4\pi\epsilon}} e^{is(x_{N+1}-x)} \exp(-X^T A X) \\ &= \frac{1}{2\pi} \int ds e^{-isx} \left[\int \frac{dx_1}{\sqrt{4\pi\epsilon}} \dots \int \frac{dx_{N+1}}{\sqrt{4\pi\epsilon}} \exp(-X^T A X + isx_{N+1}) \right] \end{aligned}$$

We write

$$(\dots) = X^T A X - F^T X, \quad F = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ +is \end{pmatrix}_{N+1}$$

and shift X to $X = Y + \gamma$:

$$\begin{aligned} X^T A X - F^T X &= (Y + \gamma)^T A (Y + \gamma) - F^T (Y + \gamma) \\ &= Y^T A Y + \underbrace{Y^T A \gamma + \gamma^T A Y}_{=0} + \gamma^T A \gamma - \underbrace{F^T Y - F^T \gamma}_{=0} \\ &= Y^T A Y + \underbrace{Y^T (2A\gamma - F)}_{=0} + [\gamma^T A \gamma - F^T \gamma] \\ &\quad \quad \quad = 0 \Rightarrow \gamma = +\frac{1}{2} A^{-1} F \end{aligned}$$

so that

$$\begin{aligned} [\dots] &= \gamma^T A \gamma - F^T \gamma = \frac{1}{4} F^T A^{-1} F + \frac{1}{2} F^T A^{-1} F = -\frac{1}{4} F^T A^{-1} F \\ &= -\frac{1}{4} (is)^2 (A^{-1})_{N+1, N+1} = +\frac{s^2}{4} (A^{-1})_{N+1, N+1} \end{aligned}$$

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which gives: (Jacobian from X to Y is $= 1$):

$$I_4^{\text{cond}} = \frac{1}{2\pi} \int_{-\infty}^{\infty} ds e^{-isx} e^{-\frac{s^2}{4} (A^{-1})_{N+1, N+1}} \times \frac{1}{\sqrt{\mathcal{D}(0)}}$$

where $\mathcal{D}(t)$ satisfies:

$$\begin{cases} \frac{d\mathcal{D}}{d\tau} = p(\tau) \mathcal{D}(\tau) & , \quad \mathcal{D}(t) = 1 \\ \frac{d\mathcal{D}}{d\tau} \Big|_{\tau=t} = 0 \end{cases}$$

Now we have $\int ds$:

$$\int_{-\infty}^{\infty} ds e^{-\alpha s^2 - isx} = \sqrt{\frac{\pi}{\alpha}} \exp\left(\frac{(-ix)^2}{4\alpha}\right) = \sqrt{\frac{\pi}{\alpha}} e^{-x^2/4\alpha}$$

where $\alpha = \frac{1}{4} (A^{-1})_{N+1, N+1}$, and, hence,

$$I_4^{\text{cond}} = \frac{1}{\sqrt{\pi \mathcal{D}(0) (A^{-1})_{N+1, N+1}}} e^{-x^2 / (A^{-1})_{N+1, N+1}}$$

• We need to calculate $(A^{-1})_{N+1, N+1}$.

We introduce a new sequence of determinants containing the first k rows and columns of A :

$$G_k = \begin{vmatrix} a_1 - 1/\varepsilon & & & \\ -1/\varepsilon & a_2 - 1/\varepsilon & & \\ & -1/\varepsilon & a_3 - 1/\varepsilon & \\ & & \ddots & \ddots \\ & & & -1/\varepsilon & a_k - 1/\varepsilon \\ & & & & -1/\varepsilon & a_k \end{vmatrix}$$

• Open along the last row:

$$G_k = a_k G_{k-1} + 1/\varepsilon \begin{vmatrix} \ddots & & 0 \\ -1/\varepsilon & a_{k-2} & 0 \\ 0 & 0 & -1/\varepsilon & -1/\varepsilon \end{vmatrix}$$

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$$= a_k G_{k-1}^1 + 1/\epsilon (-1/\epsilon) G_{k-2}^1 = a_k G_{k-1}^1 - \frac{1}{\epsilon^2} G_{k-2}^1$$

$$\text{or } G_k^1 = \left(p_k \epsilon + \frac{2}{\epsilon} \right) G_{k-1}^1 - \frac{1}{\epsilon^2} G_{k-2}^1$$

$$G_k^1 - \frac{2}{\epsilon} G_{k-1}^1 + \frac{1}{\epsilon^2} G_{k-2}^1 = p_k \epsilon G_{k-1}^1$$

~~Let $A_k = \epsilon G_k^1$~~ And we need:

$$(A^{-1})_{N+1, N+1} = \frac{(\text{Co-factor of } a_k \text{ in } A)}{\det A}$$

$$\det A \equiv G_{N+1}^1 \quad \text{co-factor} \equiv G_{N+1}^1$$

$$(A^{-1})_{N+1, N+1} = \frac{G_{N+1}^1}{G_{N+1}^1} = \frac{\epsilon^{N+1} G_{N+1}^1}{\epsilon^{N+1} G_{N+1}^1} = \frac{\tilde{G}_{N+1}^1}{\tilde{G}_{N+1}^1}$$

where

$$\epsilon^k G_k^1 = \tilde{G}_k^1 = \begin{vmatrix} a_1 \epsilon & -1 & & & \\ -1 & a_2 \epsilon & -1 & & \\ & & & & \\ & & & & \\ & & & -1 & a_{k-1} \epsilon & -1 \\ & & & -1 & a_k \epsilon \end{vmatrix}$$

$$\tilde{G}_k^1 = (a_k \epsilon) \tilde{G}_{k-1}^1 - 1 \cdot \tilde{G}_{k-2}^1 = \underbrace{a_k \epsilon}_{p_k \epsilon^2 + 2} \tilde{G}_{k-1}^1 - \tilde{G}_{k-2}^1$$

$$\hookrightarrow \boxed{\frac{d^2 \tilde{G}(\tau)}{d\tau^2} = p(\tau) \tilde{G}(\tau)}$$

~~$\tilde{G}_k^1$~~ $\Rightarrow \tilde{G}$ Note that $a_k \epsilon = p_k \epsilon^2 + 2$ for all

$k=1, \dots, N$.

Consider now $\tilde{Q}_k = \epsilon \tilde{G}_k^1$. It satisfies the same

$$\text{DE } \boxed{\frac{d^2 \tilde{Q}(\tau)}{d\tau^2} = p(\tau) \tilde{Q}(\tau)} \text{ and the boundary}$$

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conditions:

$$\tilde{Q}_N = \varepsilon(p_N \varepsilon^2 + 2) \rightarrow 0 \Rightarrow \boxed{\tilde{Q}(t) = 0}$$

and

$$\frac{1}{\varepsilon} [\tilde{Q}_N - \tilde{Q}_{N-1}] = \frac{1}{\varepsilon} [\varepsilon \tilde{G}_N - \varepsilon \tilde{G}_{N-1}] = \tilde{G}_N - \tilde{G}_{N-1}$$

$$= (p_N \varepsilon^2 + 2) - \begin{vmatrix} p_{N-1} \varepsilon^2 + 2 & -1 \\ -1 & p_N \varepsilon^2 + 2 \end{vmatrix}$$

$$= (p_N \varepsilon^2 + 2) - [(p_{N-1} \varepsilon^2 + 2)(p_N \varepsilon^2 + 2) - 1] = 2 - 4 + 1 + O(\varepsilon^2) \rightarrow -1$$

we

$$\boxed{\left. \frac{d\tilde{Q}(\tau)}{d\tau} \right|_{\tau \rightarrow t} = -1}$$

and hence

$$(A^{-1})_{N+1, N+1} \rightarrow \frac{\tilde{Q}(0)}{\tilde{Q}(0)}$$

$\hookrightarrow$

$$\boxed{I_4 = \frac{1}{\sqrt{\pi \tilde{Q}(0)}} \exp \left[-x^2 \frac{\tilde{Q}(0)}{\tilde{Q}(0)} \right]}$$

(1.1.113)